

## DPT Exercises

Exercise sheet 12

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**Exercise 1** (confidence interval for a proportion). A polling agency wants to estimate the proportion  $p$  of voters who support a particular candidate. They conduct a (uniformly!) random sample of  $n = 400$  voters and find that  $k = 220$  of them support the candidate.

- (i) Calculate the sample proportion  $\hat{p}$  of voters who support the candidate.
- (ii) Construct an approximate 95% confidence interval for the true proportion  $p$ .
- (iii) Interpret the confidence interval in the context of this problem.
- (iv) Assume the agency wants to achieve a “margin of error” of  $\pm 0.02$  for the 95% ci. The margin of error is the quantity that gets added to and subtracted from the point estimate to obtain the ci. Approximately how many voters would they need to sample? (Assume  $\hat{p}$  from the initial sample is a reasonable estimate for  $p$  for this planning.)

**Solution:**

- (i) The sample proportion is  $\hat{p} = k/n = 220/400 = 0.55$ .
- (ii) For a 95% confidence interval, the significance level is  $\alpha = 0.05$ , so  $z_{\alpha/2} = z_{0.025} \approx 1.96$ . The standard error of the sample proportion is

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.55(1-0.55)}{400}} = \sqrt{\frac{0.55 \cdot 0.45}{400}} = \sqrt{\frac{0.2475}{400}} \approx 0.02487.$$

The approximate 95% ci for  $p$  using the Wald interval is

$$\hat{p} \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.55 \pm 1.96 \cdot 0.02487.$$

So the ci is approximately (0.5, 0.6).

- (iii) If we were to repeat this polling process many times, each time selecting 400 voters, in 95% of these processes the confidence interval constructed in this way would contain the true population proportion  $p$ . Specifically, we may obtain a different confidence interval (and point estimate) in each repetition.
- (iv) The margin of error (ME) is  $ME = z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$ . We want  $ME = 0.02$ . Using  $z_{0.025} \approx 1.96$  and estimating  $p$  with  $\hat{p} = 0.55$  we simply rearrange

$$0.02 = 1.96 \sqrt{\frac{0.55(1-0.55)}{n}} \implies n = \left(\frac{1.96}{0.02}\right)^2 0.55(1-0.55) \approx 2377.$$

The agency would need to sample approximately 2377 voters to obtain the desired margin of error. (If  $p$  was unknown, one often uses  $p = 0.5$  for the most conservative (largest) sample size estimate, which would give  $n = 2401$ ). This process is akin to the process of power analysis for hypothesis testing: given some assumptions, figure out the required sample size to obtain a desired level of power at a fixed significance level.

**Exercise 2** (typos and errors). Find as many typos and errors or ambiguities and inconsistencies in the lecture notes as you can and report them (either on Moodle or directly to [niki.kilbertus@tum.de](mailto:niki.kilbertus@tum.de)).

**Exercise 3** (study and practice). Study for the exam and practice the exercises.